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A PALEOMAGNETIC STUDY ON THE GUNFLINT, MESABI, AND CUYUNA IRON RANGES IN THE LAKE SUPERIOR REGION

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ABSTRACT

One hundred and eleven samples of Animikie iron formation, partially altered iron formation, and direct-shipping iron ore were collected from the Gunflint, Mesabi, and Cuyuna Ranges in the Lake Superior region of North America. Paleomagnetic measurements were made on the samples using an astatic magnetometer, and demagnetization by an alternating magnetic field was applied to test for remanence stability. The primary iron formations—the Gunflint, Biwabik, and Deerwood iron formations from the Gunflint, Mesabi, and Cuyuna Ranges, respectively—appear to be contemporaneous with stable, primary chemical remanent magnetization (CRM) which, as indicated by the Gunflint iron formation, yields an Animikie paleomagnetic pole position of 94°W, 28°N. The remanence characteristics of the Gunflint iron formation further suggest that the iron formation was deposited at a very slow rate in a polar geographic region, and that reversals in the earth's magnetic field may have occurred during deposition. The samples from the Mesabi and Cuyuna Ranges yield a smeared distribution of remanence directions that start from Animikie—Lower Keweenawan directions for the primary iron formation samples, and end in Mesozoic—Cenozoic directions for the partially altered iron formation and soft iron ore samples. The hypothesis of ore genesis which best fits this distribution is that meteoric solutions weathered the primary Animikie iron formations during the Mesozoic—Cenozoic to form the soft, direct-shipping iron ore deposits. The oxidation of the primary iron minerals by the solutions resulted in a stable, secondary CRM in the ore samples. The partially altered iron formations probably have intermediate remanence directions produced by the combination of primary and secondary remanence components.

INTRODUCTION

SEVERAL investigators have suggested that the Gunflint, Biwabik, and Deerwood iron formations of the Gunflint, Mesabi, and Cuyuna Ranges, respectively, were deposited contemporaneously, and that the soft, direct-shipping iron ores of the Mesabi and Cuyuna Ranges were formed by the same genetic process at the same time. These Ranges are located in the Lake Superior region of North America as shown in Figure 1. There are significant geologic differences between the Ranges as noted in Table 1. These differences have led the investigators to propose several different hypothses of ore genesis, and they provide the necessary elements to distinguish between different hypotheses by using paleomagnetism.

Relatively few paleomagnetic studies have been reported on iron formations and iron ores, and none have been reported on those of the Lake Superior

TABLE 1

COMPARISON OF SOME GEOLOGIC FEATURES OF THE IRON FORMATIONS

Feature	Iron formation		
	Gunflint	Biwabik	Deerwood
Degree of tectonic deformation	Flat lying and undeformed	Very gently folded	Moderately folded into synclines and anticlines
Degree of metamorphism	Unmetamorphosed	Unmetamorphosed	Slightly metamorphosed to the chlorite facies
Degrees of alteration to "soft" iron ore	Unaltered; no ore deposits	Extensively altered to form large deposits	Extensively altered to form small deposits

region. Nairn (32), Zavoiski and Krutihovskaya (48), and Vlassov and Aparin (43) have reported on iron formations. Howell et al. (19), Carmichael (2), Hargraves (18), Gross and Strangway (10), and R. L. DuBois (5) have reported on iron deposits. Gross and Strangway (10) in 1961 were the first to use paleomagnetism primarily to investigate the genesis of iron deposits.

More detailed discussions on the equipment and experimental techniques used in this study are given in a thesis by the author (39).

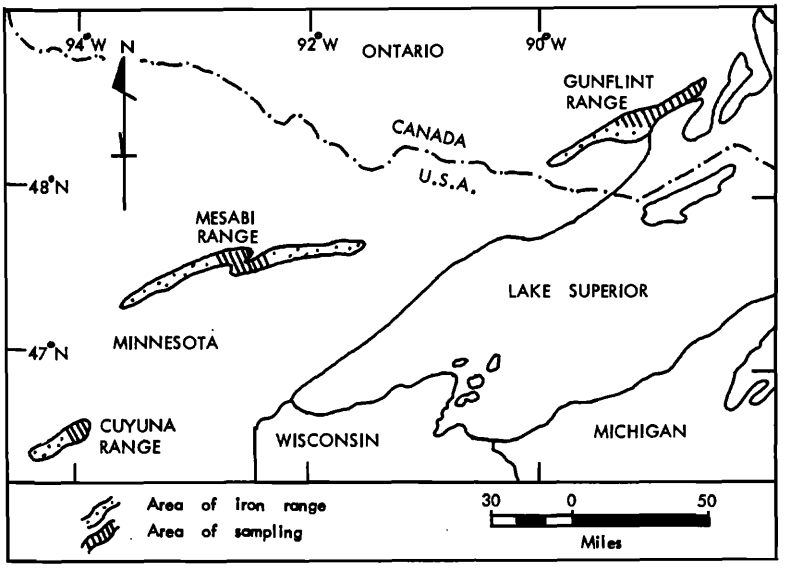


FIG. 1 Location map showing the Gunflint, Mesabi, and Cuyuna Iron Ranges of North America.

GEOLOGY

Gunflint Range.—The general geology of the Gunflint Range (9, 30, 31, 40) to the east of Kakabeka Falls is shown in Figure 2, and its stratigraphic sequence is given in Table 2.

The Gunflint iron formation of Animikie age unconformably overlies a basement complex of Early Precambrian rocks. The iron formation is subdivided into Lower and Upper members. The Lower member contains a

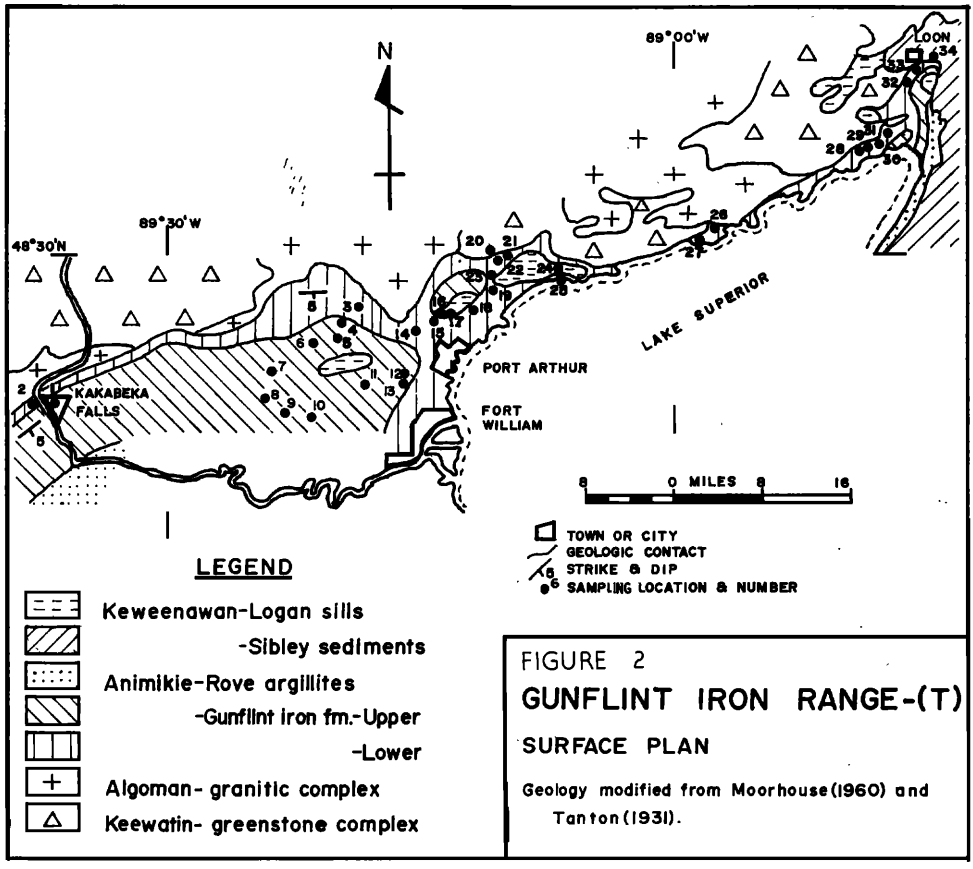


FIGURE 2
GUNFLINT IRON RANGE-(T)
SURFACE PLAN
Geology modified from Moorhouse(1960) and Tanton(1931).

basal conglomerate unit (0 to 4 feet thick), a central banded chert-carbonate unit (20 to 60 feet thick), and an uppermost argillite-tuff unit (20 to 105 feet thick). The central banded chert-carbonate unit contains iron formation of the carbonate facies and hematite-banded oxide facies in the classification proposed by James (21). The Upper member varies between 100 and 150 feet in thickness. It contains iron formation of the carbonate facies with minor zones of hematite-banded oxide facies and granular silicate facies. The Gunflint iron formation is conformably overlain by the Animikian Rove

argillites (about 2,000 feet thick) which are unconformably overlain in turn by the Keweenawan Sibley clastic and chemical sediments (200 to 400 feet thick). The Logan Sills (up to 2,000 feet thick) and a few narrow diabase dikes of Keweenawan age have been intruded into the iron formation. Except for minor thermal metamorphism within a very few feet of these intrusives, the iron formation is unmetamorphosed as indicated by the preservation of delicate organic structures. The iron formation dips gently to the south at 0° to 12° and it is undeformed except for rare small normal faults with less than 40 foot displacements and associated drag folds. No commercial iron

TABLE 2

CORRELATION CHART OF THE ROCK FORMATIONS OF THE IRON RANGES

ERA	PERIOD - SYSTEM		GUNFLINT RANGE	MESABI RANGE	CUYUNA RANGE
Mesozoic	Cretaceous			Coleraine clastic sediments	Coleraine clastic sediments
	Grenville Orogeny (1.0 b.y.)*			unconformity	unconformity
Late Precambrian	Keeweenawan	Middle	Logan sills (basic intrusives)	Duluth gabbro (basic intrusives	Chlorite shist (?)
			intrusive contacts	intrusive & conformable contacts	intrusive contacts (?)
		Lower	Sibley clastic sediments & limestone		
	Penokean Orogeny (1.7 b.y.)		angular unconformity		
	Animikie	Upper	Rove argillite	Virginia slates	Upper argillites & slates
				paraconformity	
Middle Precambrian		Middle	Gunflint iron formation	Biwabik iron formation	Deerwood iron formation
				Pokegama quartzite	Lower sericitic argillite & siltstones
			unconformity	unconformity	
	Algoman Orogeny (2.5 b.y.) (Algoman-type)		acidic intrusive complex	Giants Range granites	
			intrusive contacts	intrusive contacts	
	Temiskamian (Temiskaming-type)				
Early Precambrian	Laurentian Orogeny			Knife Lake slates	
				unconformity	
	Ontarian (Keewatin-type)		basic extrusive complex	Keewatin greenstones	

* b.y. - billion years (10^9)

ore deposits have been found on the Gunflint Range. The iron formation was sampled to test the remanence characteristics of unaltered iron formation.

Mesabi Range.—The general geology of the central Mesabi Range (8, 16, 24, 42, 44, 45, 46) is shown in Figure 3 and its stratigraphic sequence is given in Table 2.

The Biwabik iron formation of Animikie age unconformably overlies a basement complex of Early Precambrian rocks and conformably overlies the Pokegama quartzite of Animikie age where the latter is present. The Biwabik iron formation is subdivided stratigraphically from bottom to top into the Lower Cherty, Lower Slaty, Upper Cherty and Upper Slaty members. More detailed descriptions of the lowermost three members are given in Table 3. The iron formation is unmetamorphosed and contains evidence of Precambrian organic structures. The iron formation is paraconformably overlain by the Animikian Virginia argillites which are unconformably over-

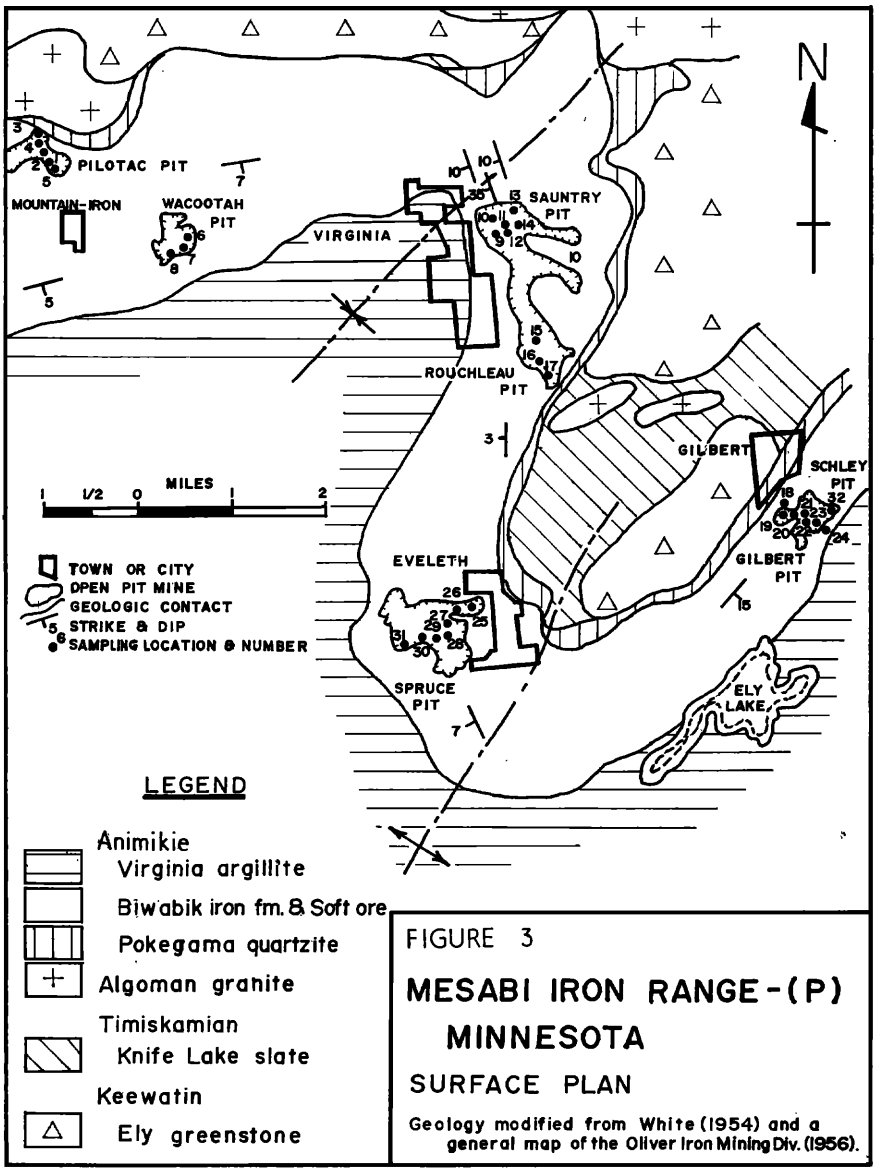


FIGURE 3

MESABI IRON RANGE-(P) MINNESOTA

SURFACE PLAN

Geology modified from White (1954) and a general map of the Oliver Iron Mining Div. (1956).

lain in turn by Keweenawan clastic sediments and igneous extrusives. Some small igneous intrusives of Keweenawan age cut the iron formation. Finally, all the Precambrian rocks are unconformably overlain by the Coleraine clastic sediments of Cretaceous age.

The Animikie sediments of the Mesabi Range (Fig. 3) strike at N70°E and dip at 6° to 12° to the southeast except around the Virginia syncline

and Eveleth anticline. The sediments are cut by some minor gravity faults with displacements that rarely exceed 50 feet and associated minor drag folds.

Two types of iron ore are mined on the central Mesabi Range from within the Biwabik iron formation. The first, concentrating ore, contains coarse-grained magnetite in iron formation of the magnetite-banded oxide facies. The second, direct-shipping ore, contains soft martite and goethite with minor magnetite and specularite, and quartz as the major gangue mineral. The direct-shipping or soft ore is porous and friable, and it contains numerous slump structures. The soft ore deposits are found in areas of fracture such as anticlinal crests, and almost entirely within 500 feet of the surface. The ore bottoms on Pokegama quartzite or in unoxidized iron formation. The occurrence of the ore is unrelated to the Keweenawan dikes.

Cuyuna Range.—The general geology of the Cuyuna Range (8, 12, 17, 21, 22, 34, 35, 36, 42, 47) and of the sampled open pit mines are illustrated in Figures 4 and 5. The stratigraphic sequence is given in Table 2.

The Animikie is represented by the Lower argillites (about 2,000 feet thick) which are conformably overlain by the Deerwood iron formation (45 to 500 feet thick) and then by the Upper argillites and slates (over 2,000 feet thick). The iron formation contains a lower thinly-bedded member of the

TABLE 3

STRATIGRAPHIC DESCRIPTION OF THE LOWERMOST THREE MEMBERS OF THE BIWABIK IRON FORMATION

Member	Unit	Thickness in feet	Description
Lower Cherty	I	30 - 50	Hematite-banded oxide facies of iron formation with inter-bedded sandstones, conglomerates, and algal reef structures.
	II	40 - 70	Hematite and magnetite-banded oxide facies of iron formation in regular bands.
	III	80 - 140	Magnetite-banded oxide facies of iron formation in irregular bands with some brown and green slaty layers.
	IV	20 - 80	Magnetite-banded oxide facies of iron formation in irregular bands mixed with the granular silicate facies of iron formation in regular bands.
	V	20 - 50	Granular silicate facies of iron formation in regular but poorly defined bands.
Lower Slaty		10 - 70	Black, brown, and green slate which grades into the granular silicate and carbonate facies of iron formation
Upper Cherty		260 - 320	Non-granular and granular silicate facies of iron formation which grade into the magnetite-banded oxide facies of iron formation near the top of the unit.

carbonate facies, a thin central member of argillaceous "paint rock," and an upper thickly-bedded member of the granular silicate facies. The iron formation belongs to the chlorite facies of thermal metamorphic grades. The Chlorite shists may be either metamorphosed igneous intrusives of Keweenawan age or metamorphosed sediments of Animikie age as both igneous and sedimentary textures are reported.

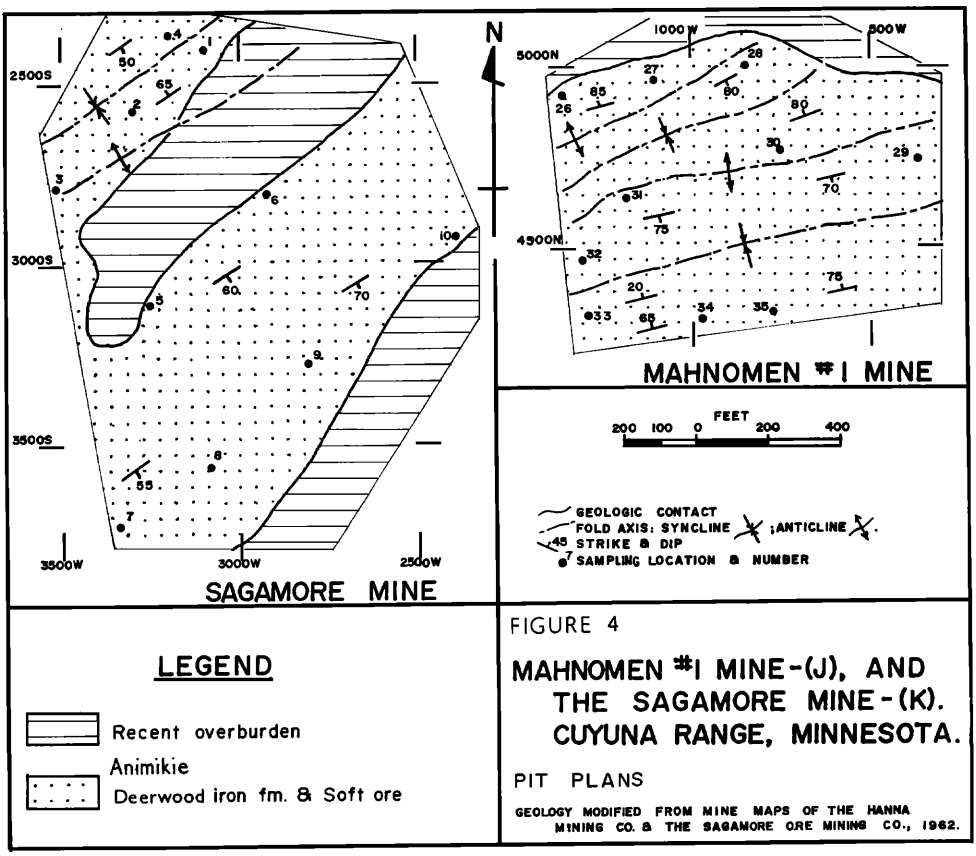


FIGURE 4

MAHNOMEN #1 MINE-(J), AND THE SAGAMORE MINE-(K). CUYUNA RANGE, MINNESOTA.

PIT PLANS

GEOLOGY MODIFIED FROM MINE MAPS OF THE HANNA MINING CO. & THE SAGAMORE ORE MINING CO., 1962.

The regional structure is a series of alternating synclines and anticlines whose axial planes trend N65°E, whose axes plunge gently in both directions, and whose limbs are overturned with southerly dips (Fig. 4). Numerous minor synclines and anticlines are superimposed on these regional folds (Figs. 4, 5).

There are three types of soft, direct-shipping iron ore that have been mined from within the Biwabik iron formation of the Cuyuna Range: "brown ore" contains almost no manganese; average "manganiferous ore" contains up to 8 percent manganese; and "black ore" contains up to 18 percent manganese. The major iron minerals in the ores are goethite with minor hematite and mag-

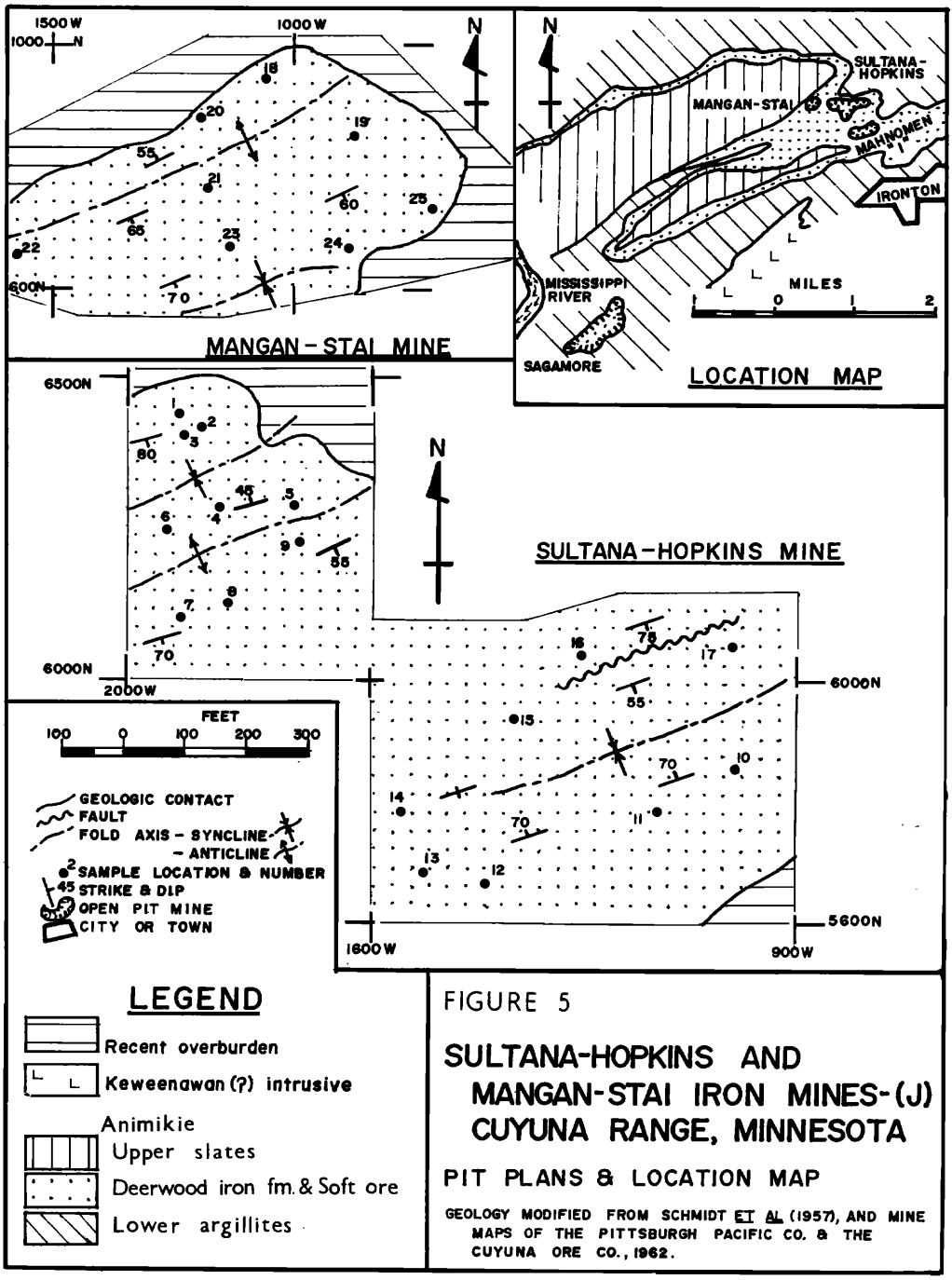


FIGURE 5

SULTANA-HOPKINS AND MANGAN-STAI IRON MINES-(J) CUYUNA RANGE, MINNESOTA

PIT PLANS & LOCATION MAP

GEOLOGY MODIFIED FROM SCHMIDT ET AL (1957), AND MINE MAPS OF THE PITTSBURGH PACIFIC CO. & THE CUYUNA ORE CO., 1962.

TABLE 4

STRATIGRAPHIC CORRELATION BETWEEN THE IRON FORMATIONS OF THE GUNFLINT, MESABI, AND CUYUNA IRON RANGES

GUNFLINT RANGE	MESABI RANGE	CUYUNA RANGE
	BIWABIK IRON FORMATION Upper Slaty member - slaty taconite	UPPER ARGILLITES AND SLATES with tuffs and flows in basal 200 feet
GUNFLINT IRON FORMATION Upper member - carbonate and granular silicate facies	Upper Cherty member - granular silicate and oxide facies	DEERWOOD IRON FORMATION Upper member - granular silicate facies
Lower member - upper argillite - tuff unit	Lower Slaty member - slaty taconite	Central member - argillaceous "paint rock"
- middle carbonate facies unit	Lower Cherty member - oxide facies of units II, III, IV & V	Lower member - carbonate facies
- lower conglomerate unit	- basal conglomerate of unit I	

netite. The major manganese mineral is psilomelane with minor pyrolusite. The most abundant impurities are titanium and phosphorus. The ores are soft and porous with their iron minerals commonly showing replacement textures. The ores show the original bedding laminations, which are preserved as aluminous and argillaceous laminations. Slump structures are not common in the ore. The ore is confined to a zone within 300 feet of the surface on both anticlinal and synclinal structures. The ore deposits bottom on either unoxidized iron formation or argillaceous layers of sediment.

Correlation between the Iron Ranges.—All three notable attempts (25, 29, 42) to correlate the geology between the Lake Superior iron districts on geological arguments agreed that the Gunflint, Biwabik and Deerwood iron formations are contemporaneous. Their contemporaneity has been reaffirmed by Goldich et al. (8) on the basis of more recent geologic evidence and of radioactive age determinations. The age determinations (8, 27, 28, 37) suggest a minimum age of 1.7 billion (10^9) years for the Animikie iron formations which corresponds to the Penokean orogeny on the basis of six K-A determinations from the Deerwood iron formation and the Virginia slates. On the basis of about one hundred K-A and Sr-Rb determinations from rocks of the pre-Animikie basement complex, a maximum age of 2.3 billion years is suggested for the iron formations. Several correlations between members of the iron formations have been suggested of which two are indicated in Table 4 (7, 12, 40).

Genesis of the Iron Formations and Ore Deposits.—Van Hise and Leith (42) in 1911 strongly advocated the hypothesis that the iron formations were deposited as chemical sediments with their present mineralogy and textures during quiescent intervals of volcanicity. James (21) in 1954 subdivided the Precambrian iron formations into six lithologic facies or subfacies. He pro-

posed that the different facies were produced by changing sedimentary conditions at different locations in the restricted marine depositional basin. The location in the basin controls the clastic sedimentary content of the iron formation as for example, coarse grains will deposit near the basin's edges whereas clay-sized grains will deposit near the basin's center, and the chemical sedimentary content through varying Eh-pH conditions as for example, hematite and magnetite will deposit under oxidizing conditions near the basin's edges whereas pyrite and siderite will deposit under reducing conditions near the basin's center. Lepp and Goldich (26) in 1964 argued that extensive chemical transformations occurred in the primary iron formation during diagenesis to convert the original chemical sediment into the various facies of iron formation. In each hypothesis, the remanence of the iron formation will be formed when the iron formation is essentially horizontal, and it will be composed of a DRM (detrital remanent magnetization) for the clastic sediment and a CRM (chemical remanent magnetization) for the chemical sediment in the iron formation.

The arguments over the genesis of the soft, direct-shipping iron ores of the Mesabi and Cuyuna Ranges divide into two general categories: the weathering hypothesis; and the hydrothermal hypothesis. The weathering hypothesis was proposed by Leith (24) in 1903, who suggested that meteoric solutions first oxidized the primary iron minerals, siderite and magnetite, in the iron formation to hematite and martite, respectively, and then they leached the silica from the iron formation to leave residual deposits of soft hematite and martite. Van Hise and Leith (42) in 1911 suggested further that the alkaline groundwater solutions were derived from the surrounding granitic areas, and that the deposits were formed prior to the Cambrian period and for the most part during the Upper Animikie. Wolff (46) in 1915 and Zapffe (47) in 1928 have supported the hypothesis of Van Hise and Leith in discussing the Mesabi and Cuyuna Ranges, respectively. Royce(33) in 1942 favored the weathering hypothesis except that he believed that the ores were formed for the most part after the Cretaceous period or in the present erosion cycle. White (44) in 1954 when discussing the Mesabi Range concurred with the weathering hypothesis, but only went as far as to say that the ore was formed prior to the Cretaceous period. Gruner (13, 14) in 1930 and 1937 proposed the hydrothermal hypothesis. He considered that either hydrothermal solutions emanating from igneous intrusives or meteoritic solutions heated by igneous intrusives oxidized the primary iron minerals and leached the silica of the iron formation to leave residual deposits of soft iron ore. Later, Gruner (15) detailed the mineralogical changes involved in the oxidation process: magnetite yields martite, hematite remains unchanged, and the iron silicates yield mainly goethite. In discussing the Cuyuna Range, Schmidt (35) in 1959 noted that the iron formation contains a reddish brown hematite alteration that extends to depth and a yellowish brown goethite-limonite alteration that occurs near the surface. He proposed that hydrothermal solutions first oxidized the iron carbonates and silicates to hematite throughout the iron formation and that meteoric solutions then leached silica, lime, magnesia, and carbon dioxide from the altered iron formation to leave goethite-

limonite. Thus he combined the weathering and hydrothermal hypotheses. Tyler (41) in 1949 contended that thermal metamorphic reactions in the iron formation have altered the mineralogy materially to assist solution attack, and have thus controlled ore genesis. Supported by the observations of Leith (24), Grout and Broderick (11), and Royce (33), White (44) in 1954 argued against Tyler's hypothesis. The adoption of either the weathering or the hydrothermal hypothesis dictates the acceptance of a CRM in the ore as both hypotheses would produce the secondary iron ore minerals by the oxidation or chemical reaction of the primary iron minerals in the iron formation. If primary hematite survives the oxidation as Gruner (15) postulated, then a primary remanence component could survive and be present with the secondary remanence in the ore.

SAMPLING AND EXPERIMENTAL TECHNIQUES

The samples are numbered in collecting sequence for each locality, and a prefix letter precedes the number to indicate the locality. Thus "T" indicates the Gunflint Range, "P" indicates the Mesabi Range, and "J" and "K" indicate the main northeastern group of mines and the satellite Sagamore Mine, respectively, on the Cuyuna Range.

One large oriented hand sample was collected at each site. The samples were oriented by means of topographic features on mine maps or other surveyed sheets either by eye or with the aid of a Brunton compass where applicable. Potentially anomalous sites such as elevated outcrops where lightning might have struck, fault zones, joints, and blast holes, were avoided. All the samples from the Gunflint Range were collected from road cuts except for two samples from a quarry (T-12 and 13), six samples from river cuts (T-3, 10, 15, 18, 20, and 21), and five samples from outcrops (T-5, 9, 22, 27 and 34). All the samples fom the Mesabi and Cuyuna Ranges were collected in open pit mines from well below the original surface. In all 111 [1] samples were collected from iron formation or iron ore of which 29 are from the Gunflint Range, 32 are from the Mesabi Range, and 45 are from the Cuyuna Range.

From each oriented hand sample, a suitable sample for measurement was broken off using a hydraulic press. Such a suitable sample had an average volume of about 200 cc and a weight of about 600 gm. Each sample was encased in the center of a four-inch cube so that it was oriented in its original position relative to geographic coordinates. The cube was made of plaster-of-Paris of a very high purity. This method was adopted because the toughness of the iron formations and ores made coring or cutting of cubes impractical.

The remanent and induced magnetization of the samples were measured on an astatic magnetometer. Contaminants in the plaster-of-Paris and magnetic noise at the measuring site limited the sensitivity of the magnetometer to 1×10^{-7} emu/gm for a sample's magnetization in the closest acceptable position (10 cm) from the magnet system. The system of remanence measure-

[1] A further five samples (T-16, 17, 20, 24, and 34) from other rock units of the Gunflint Range are not included in this discussion.

ments was similar to that described by As (1). The remanence intensity, the remanence direction, the induced magnetization, the susceptibility, and the Koenigberger ratio of remanent to unit-field induced intensity of magnetization were calculated using standard equations from the magnetometer readings on an IBM 7094 computer. Demagnetization was done by an alternating magnetic field (i.e., AC demagnetization). The field was reduced by using a combination of variac and sample-withdrawal techniques. The samples were demagnetized along three mutually perpendicular axes. The Earth's magnetic field was nulled to within a measured two percent, which was the magnetic noise level at the site by using three sets of Helmholtz coils.

The error in the remanence directions caused by errors in orientation and measurement is estimated to be $\pm 5°$ of arc at the 95 percent confidence level. The large size of the samples in particular helps to minimize these random errors. The statistical significance determined for many groups of remanence directions when using these methods, as for example for the Gunflint iron formation in this study, suggest that this is a reasonable estimate of the error.

The NRM (natural remanent magnetization) was measured for each of the 111 samples. Nine samples were subjected to AC demagnetization in steps of 100, 200, 300, 500 and 700 oersteds (peak) and examined in polished section (Fig. 6). The remaining 102 samples were subjected to AC demagnetization at 700 oe. Twenty-five of the 102 samples that had the least stable and the least intense remanence, were subjected to AC demagnetization at 720 oe. For this 720-oe step, the sample's orientation was reversed relative to that of the 700-oe step during the entire process of demagnetization and remeasurement.

In order to apply tectonic corrections to the remanence directions, the strike and dips of sample bedding were derived from direct measurement or from recorded values on the geologic maps. The probable error in the bedding orientation is thought to be less than 5° of arc for 95 percent of the samples. The remanence directions were composed into groups on the basis of geologic considerations. For each group, the mean remanence directions with its semivertical angle of the circular standard deviation, its precision parameter, and its corresponding paleomagnetic pole positions with the semiaxes of the oval computed from the circular standard of deviation, were calculated on the computer following the equations developed by Fisher (6). Because the Gunflint and Biwabik iron formations are almost undeformed, errors in their remanence directions caused by structural rotation for which a tectonic cannot be applied are probably only a few degrees of arc at most, whereas for the more highly deformed Deerwood iron formation they might be as much as 20° of arc. These estimates appear consistent with the results.

DISCUSSION OF RESULTS

The orientation of the Earth's magnetic field has changed throughout time relative to the North American continent. This change is generally explained by either polar wander, i.e., the movement of the Earth's field, or continental drift, i.e., movement of the continents, or by a combination of both processes

(20). Thus the pole position of the Earth's field has traced a unique path relative to the North American continent back through time. Since the remanence direction formed in a sample reflects the orientation of the Earth's field at the site at the time of remanence formation, it is possible to calculate the location of the paleomagnetic polar path by measuring the remanence directions in rocks of known age, and by making the not unreasonable assumption that the Earth's field has approximated an axial dipole throughout geologic time. The path for North America is shown in Figure 7 as compiled by Irving (20), P. M. DuBois (4) and Strangway (37). By using the reverse procedure, the age of the remanence in samples can be estimated from their measured mean direction. The path of the remanence directions through time for the Lake Superior region is shown on Figure 8. The path could be thought of as a band because an individual remanence direction can deviate by up to 20° from the mean direction shown by the path for its age due to secular variations in the Earth's field which have a periodicity of a few hundred years (3). Also individual directions may be oppositely directed, or reversed through 180° of arc, due either to some mineralogic process of self-reversal, or more likely to periodic reversals of polarity in the Earth's magnetic field.

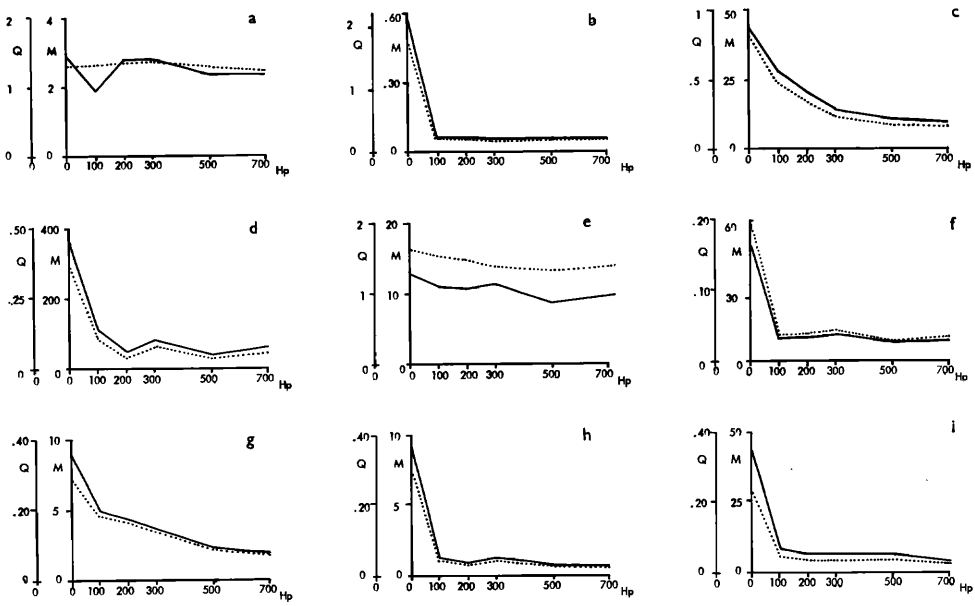


FIG. 6. AC demagnetizing curves for samples from the Gunflint, Mesabi and Cuyuna Iron Ranges. The symbols are: G, the ratio of the remanent to the unit-field induced intensity of magnetization; M, the intensity of the remanence of the sample in (emu $\times 10^{-5}$/gm); and Hp, the peak intensity of the AC demagnetizing field in oersteds. On the graphs the ratio, Q is shown by a dotted line and the intensity, M by a solid line. The samples are: a) T-22; b) T-8; c) P-14; d) P-5; e) P-9; f) J-3; g) J-5; h) J-21; and i) K-10 which are described mineralogically in Appendix A.

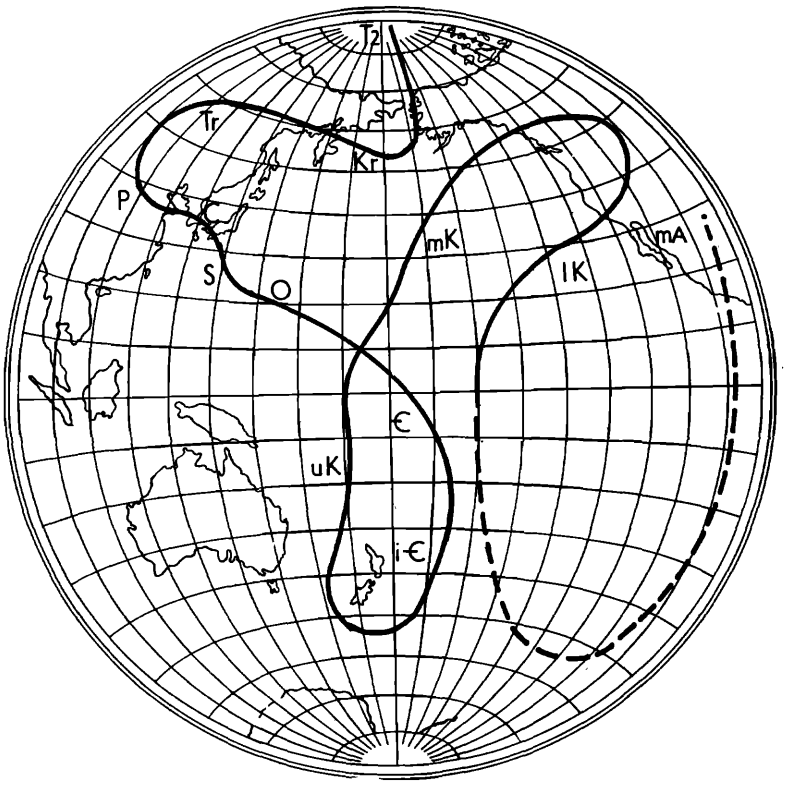


FIG. 7. Virtual paleomagnetic pole path for North America. The solid line indicates the portion of the path that is fairly well established and the discontinuous line indicates the portion not well established. The symbols are: T2, Tertiary; Kr, Cretaceous; Tr, Triassic; P, Permian; C, Carboniferous; S, Silurian; O, Ordovician; $\in$, Cambrian; i$\in$, Infra-Cambrian; uK, Upper Keweenawan; mK, Middle Keweenawan; lK, Lower Keweenawan, and mA, Middle Animikie. The path for the Tertiary to the Cambrian was interpreted by Irving (20), from the Infra-Cambrian to the Lower Keweenawan was interpreted by P. M. DuBois (4) and Strangway (38), and the Middle Animikie position is represented by the Gunflint iron formation reported in this paper.

Exclusion of Samples.—As is done in most paleomagnetic studies, it was found necessary to exclude some samples because of their unsuitable magnetic characteristics. Two samples (T-23 and 32) were excluded because their remanence intensities were below the limit of instrumental accuracy. Eighteen samples (T-1, 4, 7, 9, 18 and 19; P-1, 2, 3, 4, 17, 18, 22, 23, 24 and 31; J-17 and 34) were excluded because they showed evidence of magnetic instability by failing one or more of the following tests:

a) their ratio (Q) of the remanence intensity after demagnetization (Ms) to the induced intensity of a unit field (Mi) fell below one-hundredth (i.e., $Q = Ms/Mi < 0.01$). In samples with such a low ratio, the remanence measurement is liable to be inaccurate because the response of the magnetometer

is so much greater to the moment induced by the Earth's field than to the remanent moment that the latter becomes masked;

b) their remanence direction changed by over 60° of arc upon demagnetization at 720 oe with the sample's orientation reversed when compared to the direction of the 700 oe step; or,

c) their remanence direction was observed to be changing during measurement.

Two additional samples (T-28; P-30) were excluded because they gave anomalous remanence directions which diverge by 70° of arc from the other

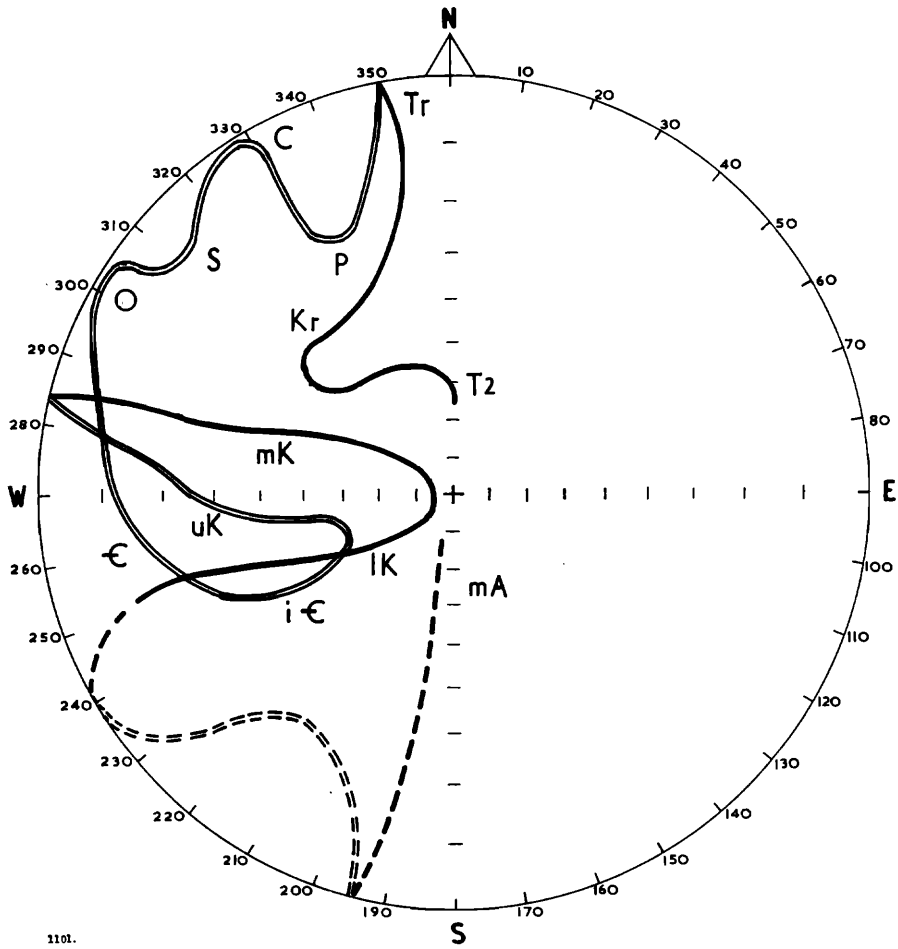


FIG. 8. Paleomagnetic path of the remanence directions for the Lake Superior area of North America. The continuous solid (open) line indicates the portion of the path that is fairly well established and the discontinuous solid (open) line indicates the portion that is not well established. Solid (open) portions of the line indicate a downwardly (upwardly) directed remanence direction. The symbols are the same as for Figure 7.

TABLE 5

SUMMARY OF THE MAGNETIC MEASUREMENTS OF SAMPLES FROM THE GUNFLINT (LONG. 89.2°W; LAT. 48.5°N), THE MESABI (LONG. 92.5°W; LAT. 47.5°N), AND THE CUYUNA (LONG. 94.0°W; LAT. 46.5°N) IRON RANGES. THE SYMBOLS ARE:,

SAMPLE GROUP	N		X	Ms	Mn/Ms	Mean Remanence vector					Pole Position			
						R	Dm	Im	α63	K	Dp	Ip	dp	dm
Gunflint iron formation														
- Lower I	4	C				3.92	5	80	7	39				
- Lower II	4	C				3.80	30	-79	12	15				
- Lower III	4	C				3.88	173	65	9	26				
- Lower	13	C*	.36 - 212	.016 - 9.5	1 - 7	12.18	174	81	6	14.7	81°W	31°N	11	12
- Upper	7	C	1.3 - 15	.055 - .77	2 - 10**	6.54	211	73	9	13.1	105°W	21°N	15	16
- Overall	20	C*				18.5	192	79	5	14.0	94°W	28°N	9	10
Biwabik iron formation														
- Lower Cherty	13		1.51 - 1417	.51 - 117	1.5 - 4									
- Lower Slaty	3		1.2 - 2994	1.2 - 45	1 - 6									
- Upper Cherty	5		12 - 258	4.4 - 23	1.3 - 4									
- Excluded	11		6.0 - 2164	16 - 75	4 - 25									
Deerwood iron formation														
- Sultana-Hopkins	16		.74 - 955	.33 - 25	1 - 13									
- Mangan-Stai	8		.18 - 180	.10 - 4.0	1 - 11									
- Mahnomen No. 1	9		.20 - 134	.025 - 3.9	1 - 5									
- Sagamore	10		.96 - 323	.085 - 7.2	1 - 10									

*Includes samples from Lower II that have been reversed through 180° of arc.

**Includes one sample with a ratio of 60 whose stability was demonstrated by AC demagnetization with the sample orientation reversed.

N—The number of samples represented in the group or the population of the remanence directions,

C—indicates that the remanence directions have been corrected for tectonic deformation prior to calculation of the mean remanence direction and pole position,

X—the range of susceptibilities in (emu $\times 10^{-6}$/gm),

Ms—the range of intensities of the stable remanences in (emu $\times 10^{-6}$/gm) after AC demagnetization at 700 oersteds (peak),

Mn / Ms—the range of ratios of NRM intensity to Ms intensity which roughly indicates the ratios of unstable (Mn-Ms) to stable (Ms) remanence,

R—the length of the mean remanence vector with each of the N remanence directions considered as a unit vector,

Dm—the declination of the mean remanence vector in degrees reckoned clockwise from 0°,

Im—the inclination of the mean remanence vector in degrees reckoned from the horizontal plane and considered positive (negative) if directed downwards (upwards),

$\alpha 63$—the semi-angle of the circular standard deviation, in degrees, of the cluster of remanence directions,

k—the precision parameter of the cluster of remanence directions,

Dp(Ip)—the longitude (latitude) of the virtual paleomagnetic pole position in degrees,

dp(dm)—the semi-axis of the oval of confidence derived from the circular standard deviation perpendicular to (along) the meridian between the sampling site and the pole position in degrees about the pole position.

directions of their respective geologic groups. Thus 22 of the 111 samples were excluded for various reasons leaving 89 samples that are considered valid.

Gunflint Range.—The samples from the Gunflint Range were collected from the flat-lying Gunflint iron formation along an exposed length of 85 miles (Fig. 2).

The thirteen valid samples from the Lower Gunflint iron formation were collected from the central banded chert-carbonate unit. These samples yield a broad range of magnetic characteristics (Table 5), and in general they have stable remanent components of low intensity (Fig. 6a), which may be indicative of CRM. After tectonic correction for the very low bedding angle, the cluster of remanence directions of the samples has a precision parameter

value of 13.3 (Fig. 9a). Such a value indicates that a moderately high statistical significance can be attached to this geological grouping of directions. The samples, excluding for the moment one (T-15) which comes from within four inches of a Logan sill, can be subdivided into three groups: Group 1 (T-3, 21, 22, and 27) comes from the lower portion of the central banded

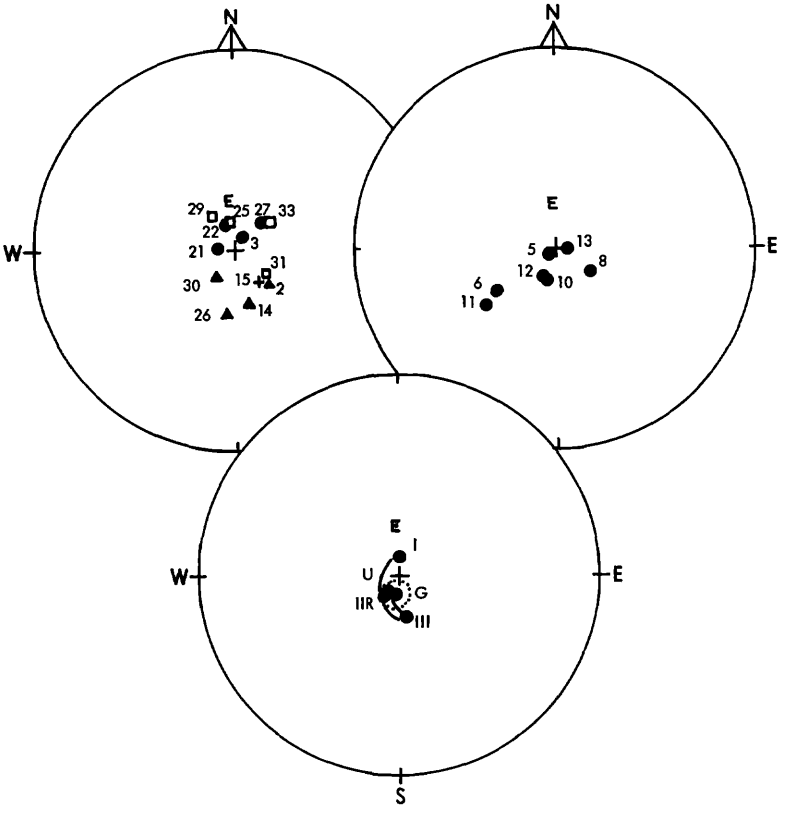


FIG. 9. Remanence directions for samples from the Gunflint iron formation after tectonic corrections were made by rotating the dip to the horizontal about the strike axis: a) Lower member showing the units Lower I (circle), Lower II (square), Lower III (triangle), and the sample (cross) near the Logan Sill contact. b) Upper member (circle). c) Gunflint iron formation showing the mean directions for Lower I (I), Lower II (IIR) after an 180° reversal, Lower III (III), Upper (U), and entire (G) iron formation. The solid line indicates the movement of the mean remanence direction during deposition of the Gunflint iron formation. The sample numbers have the prefix letter "T." Conventions: The stereonet plots are Wolfe equal angle polar projections looking down. The declination of the remanence directions is reckoned clockwise from the "North" position. The inclination is reckoned from the primitive horizon and the direction is indicated by solid (open) symbols if it is positive (negative) or downwardly (upwardly) directed. The direction of the present Earth's magnetic field in the Lake Superior region is indicated by the symbol "E." The circular standard deviation is indicated by a dashed line.

chert-carbonate unit, Group II (T-25, 29, 31 and 33) with the reversed remanence direction comes from the middle portion, and Group III (T-2, 14, 26 and 30) comes from the upper portion of the central banded chert-carbonate unit. The precision parameters for the prefolding remanence directions of the three groups, I, II, and III are: 38, 26, and 19, respectively. The mean remanence directions for these three groups (Fig. 9c) suggest that either secular variation or polar wander, and a reversal in the Earth's magnetic field occurred during the deposition of the central banded chert-carbonate unit. The one sample (T-15) which is located immediately adjacent to the Logan sill, yields a remanence direction that is very similar to those from Group III (Fig. 9a) and that differs by 65° of arc to the adjacent sill direction. This difference suggests that the temperature at the contact in the iron formation may not have exceeded 675° C, the hematite Curie Point, during intrusion.

The seven valid samples from the Upper Gunflint iron formation have a fairly restricted range of magnetic characteristics (Table 5). In general the samples have stable remanent components of low intensity, which are probably CRM and on which moderate unstable components have been superimposed (Fig. 6b). The precision parameter for the group is a statistically significant 13.1 after slight tectonic corrections are applied (Fig. 9b). The mean remanence direction for the Upper Gunflint continues the trend retraced by the mean directions for the Lower Gunflint as shown in Figure 9c.

The mean remanence direction for all the samples from the Gunflint iron formation (Fig. 9c) provides a reliable paleomagnetic pole position for the middle Animikie of North America. The stability of the remanence direction has been demonstrated: first, by subjecting the samples to AC demagnetization; second, by obtaining a statistically significant precision parameter of greater than 3.0 for the group; third, by obtaining a mean remanence direction which is divergent from the present earth's magnetic field at the site; and fourth, by obtaining exactly reversed remanence directions in the group. These paleomagnetic features when combined with the geologic features of this sedimentary horizon which is undeformed, unmetamorphosed, and unaltered, strongly support the conclusion that the remanence of the Gunflint iron formation was acquired at the time of deposition as a primary CRM.

Mesabi Range.—The twenty-one valid samples from the Mesabi Range were collected from around the Virginia syncline-Eveleth anticline, and along a strike length of 18 miles as shown on Figure 3. The samples were collected from iron formation, altered iron formation, and soft ore from within the Biwabik iron formation. The samples have the following stratigraphic distribution within the iron formation: Lower Cherty I (P-25 and 32), II (P-19 and 26), III (P-20 and 27), III or IV (P-13 and 14), IV (P-15, 21, and 28), and V (P-16 and 29); Lower Slaty (P-5, 11 and 12); and Upper Cherty (P-6, 7, 8, 9, and 10). The samples have a wide range of magnetic characteristics (Table 5). The samples with strong remanent and induced magnetic intensities contain abundant primary magnetite (Fig. 6d) and they tend to have large unstable remanence components. Those samples with weaker intensities contain either a primary iron mineral of low intensity such as hematite (Fig. 6e) or a primary iron mineral that has been mostly altered

to a secondary iron mineral such as magnetite to martite (Figs. 6f, 10a), and they tend to have quite stable remanences. Applying tectonic corrections for the fold structure to the remanence directions of the samples causes the precision parameter to decrease slightly from 8.4 to 7.2 for the Lower Cherty, from 3.1 to 2.5 for the Lower Slaty, from 12.4 to 9.2 for the Upper Cherty, and from 8.0 to 6.6 for all twenty-one samples. These slight decreases suggest that a majority of the samples acquired their remanence after folding. Here the fold test is double edged for it causes the remanence directions of the primary iron formation to coalesce on unfolding, thereby increasing the precision parameter, and of the secondary ore and altered iron formation to disperse thereby decreasing the precision parameter. The remanence directions (Fig. 11) give a smeared distribution which starts in an Animikie direction (e.g., P-5, 6, 7, 10, 19, 21, 29, and 32) similar to that found for the Gunflint iron formation (Fig. 9) and end in a Mesozoic-Cenozoic direction (e.g., P-8, 11 to 14, 16, 25 to 27) similar to those shown on Figure 8 but it does not include Upper Keweenawan or Paleozoic directions. This smeared distribution and the magnetic characteristics are consistent with the hypothesis that the Animikie remanence directions with an average remanence intensity of 732×10^{-5} emu/gm represent mostly primary iron formation and that the Mesozoic-Cenozoic remanence directions with an average intensity of 29×10^{-5} emu/gm represent mostly secondary soft-ore. This interpretation suggests that the Biwabik iron formation was formed with a primary CRM or DRM during the Animikie, that the iron formation was altered to form

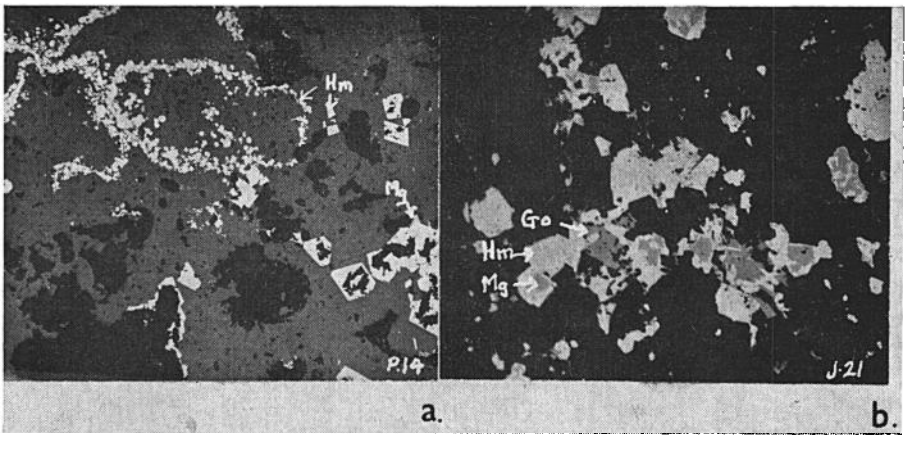


FIG. 10. Microphotographs of polished sections from samples collected within the Biwabik and Deerwood iron formations. a) Biwabik iron formation: This is sample P-14 from the Lower Cherty member which provides an example of altered iron formation. The hematite occurs as martite and as irregular clusters surrounding the silicate granules. Trace amounts of unaltered, primary magnetite were noted. × 52. b) Deerwood iron formation: This is sample J-21 from the Mangan-Stai Mine, which illustrates the mixed mineralogy of the soft ore. The blebs and crystals of magnetite have been mostly altered to hematite. The hematite is found co-existing with the goethite. × 104.

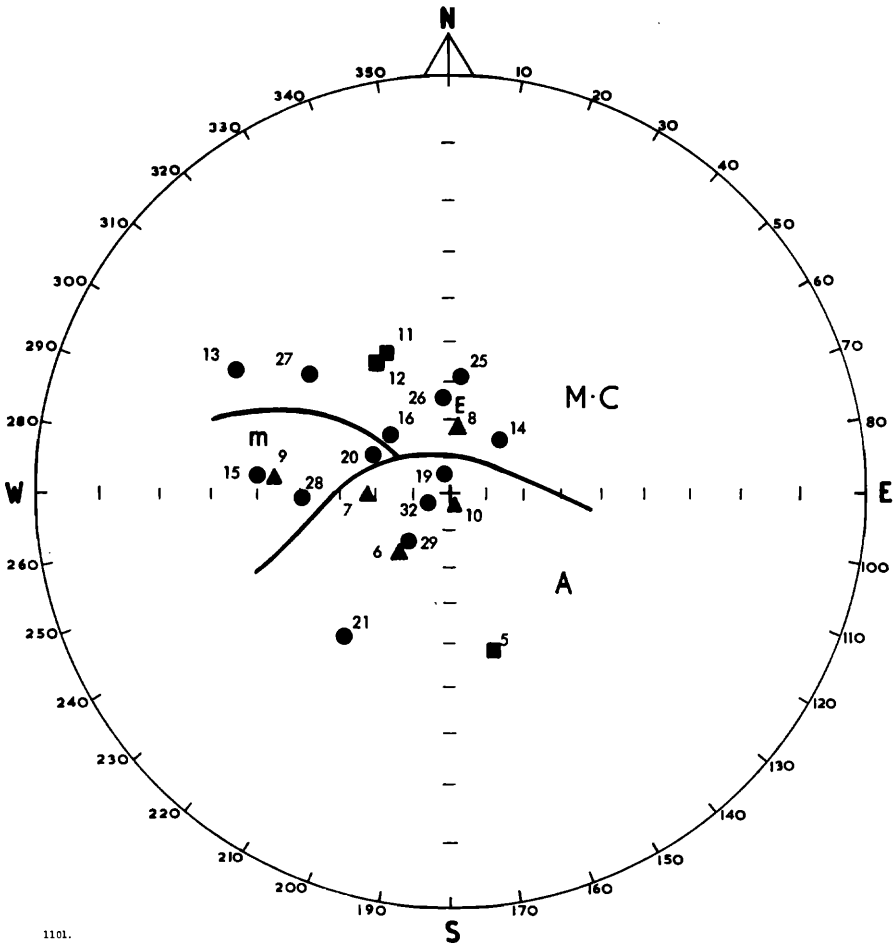


FIG. 11. Remanence directions for samples from within the Biwabik iron formation prior to tectonic correction. Different symbols indicate samples from within the Lower Cherty (circle), Lower Slaty (square), and Upper Cherty (triangle) members. The solid lines roughly separate the directions into primary Animikie (A), secondary Mesozoic-Cenozoic (MC), and mixed primary plus secondary (m) remanence groups. The sample numbers have the prefix letter "P." The conventions are the same as for Fig. 9.

soft martite-goethite ore with a CRM during the Mesozoic-Cenozoic, and that ground water was probably responsible for the alteration because there is no intrusive activity of Mesozoic-Cenozoic age anywhere in the Lake Superior region. Therefore the paleomagnetic measurements suggest that the soft ores of the Mesabi Range were formed as residual deposits of Mesozoic-Cenozoic age by the action of groundwater weathering the Biwabik iron formation of Animikie age.

Cuyuna Range.—All forty-three valid samples from the North Cuyuna Range were collected from within the Biwabik iron formation. They represent iron formation, altered iron formation, and soft ore. They were collected around anticline-syncline fold structures in the Sultana-Hopkins Mine (J-1 to 16), the Mangan-Stai Mine (J-18 to 25), and the Mahnomen No. 1 Mine (J-26 to 33, J-35), and the Sagamore Mine (K-1 to 10) (Figs. 4, 5). The first three mines are within a mile of each other, and the latter, the Sagamore Mine, is about four miles to the southwest.

The magnetic characteristics of the samples cover a broad range of values (Table 5) which reflect their mixed ferromagnetic mineralogy as shown on Figure 10b and in the descriptions for Figures 6f, 6g, 6h, and 6i. The magnetic characteristics of the samples from each of the four mines are very similar. In general, the samples have very stable remanences with quite weak intensities after demagnetization which is illustrated by the similar remanence intensities between the 100 oe and 700 oe steps on the demagnetizing curves (Figs. 6f, 6g, 6h, 6i). Unfolding causes the precision parameter for the remanence directions to decrease from a statistically significant 4.1 (Fig. 12) to an insignificant 1.3, which strongly suggest that the remanence was acquired for the most part after folding. This suggestion is supported by the observation that the mean remanence direction of twenty-five samples from southerly dipping beds differs by only 18° of arc from that of seventeen samples from northerly dipping beds, whereas the average bedding dips are about 90° apart, which indicates that only a very small primary component of remanence can be present relative to a large postfolding component. This small primary component appears to dip into the southwest quadrant, which is similar to the Animikie direction found for the Gunflint iron formation. The postfolding remanence directions have a smeared distribution (Fig. 12) which starts in Animikie remanence directions as shown in Figure 9 for the Gunflint iron formation, but much dispersed by tectonic folding, and ends in a Mesozoic-Cenozoic direction with a concentration of directions near the Earth's present dipole field. This smeared distribution is consistent with the hypothesis that the primary remanence directions of the Biwabik iron formation were formed during the Animikie and then dispersed by folding. The folding probably occurred during the Penokean orogeny. During the Mesozoic and especially during the Cenozoic, some of the iron formation was then altered to form the soft ore and the oxidized iron formation with secondary remanence directions. This conclusion is supported by the observation that the 22 samples in the Animikie stereonet area of Figure 12 have an average remanence intensity of 3.8×10^{-5} emu/gm, and the 21 samples in the Mesozoic-Cenozoic have an average remanence intensity of 1.8×10^{-5} emu/gm.

The samples from the Sagamore Mine (K) are "brown ore" whereas the samples from the other three mines (J) are average "manganiferous ore." Schmidt and Dutton (36) consider that the "brown ore" and the "manganiferous ore" represent different facies of the same iron-bearing horizon. Schmidt (35) further suggests that the limonite-goethite of the "brown ore" was produced by groundwater solutions during weathering, and that the hematite of the "manganiferous ore" was produced by the action of hydro-

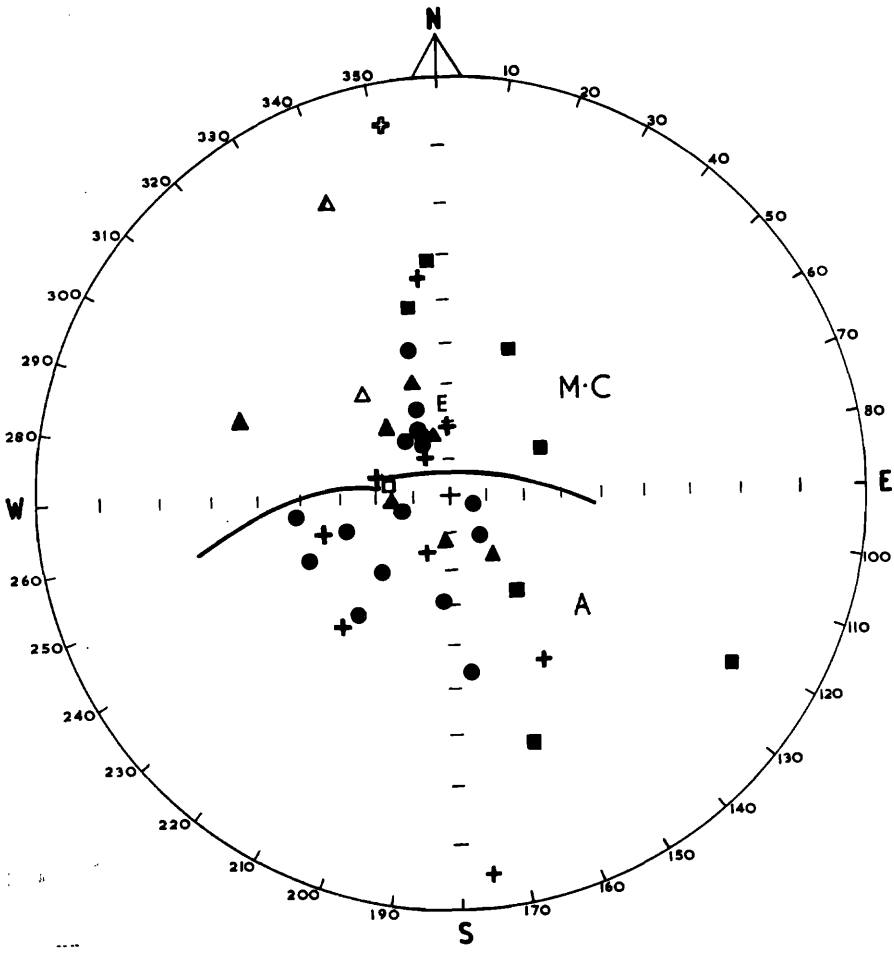


FIG. 12. Remanence directions for samples from within the Deerwood iron formation prior to tectonic correction. Different symbols indicate samples from the Sultana-Hopkins (circle), Mangan-Stai (square), Mahnomen No. 1 (triangle), and Sagamore (cross) Mines. The solid line roughly separates the directions into primary Animikie (A) and secondary Mesozoic-Cenozoic (MC) remanence groups. The conventions are the same as for Fig. 9.

thermal solutions. Zapffe (47) believes that the "brown ores" were derived from the basal iron formation and that the "manganiferous ores" were derived from the upper iron formation, some 2,000 feet higher in the stratigraphic sequence. Because the mean remanence direction prior to applying tectonic corrections of the samples from the Sagamore Mine differs by only four degrees of arc from that of the remainder of the samples, and because of the general similarity in the magnetic characteristics of both groups of samples (Table 5), it is more likely that both ores were formed contemporaneously

from contemporaneous primary iron formation which favors the interpretation of Schmidt and Dutton.

This paleomanetic data suggests that the soft ores of the Cuyuna Range were formed as residual deposits of limonite-goethite-hematite-psilomelane since the Mesozoic by the oxidizing and leaching action of groundwater on Animikie iron formation during weathering as postulated by Royce (33).

CONCLUSIONS

1. Iron formations and iron ores that are similar to those of the Gunflint, Mesabi, and Cuyuna Ranges are suitable materials for paleomagnetic study on their ore genesis.

2. The remanence directions of the Gunflint iron formation are thought to be stable, and the product of a primary CRM. The paleomagnetic pole position of 95°W, 28°N that was calculated from the mean remanence direction is considered to be a reliable pole position for the Middle Animikie.

3. The pole position for the Gunflint iron formation indicates that the iron formation was deposited in a polar geographic region which suggests that the iron formation was deposited in a cool polar climate, or that the Earth's climatic belts have changed since the Middle Animikie.

4. The progressive change in the remanence direction from the base of the Lower Gunflint through the Upper Gunflint iron formation as shown on Figure 9c, reflects a progressive change in the Earth's magnetic field. This change could be due to either secular variation, in which case the iron formation was very rapidly deposited in a few hundred years, or to polar wander in which case the iron formation was very slowly deposited over tens of millions of years. The presence of the reversed remanence directions in Group II favors the latter alternative because reversals in the Earth's magnetic field occur over periods of at least tens of thousands of years.

5. The presence of reversed remanence directions in the Gunflint iron formation confirms the presence of reversals in the Animikie rocks and probably indicates that the Earth's field changed polarity during the period.

6. The smeared distribution of remanence directions found in the samples from the Mesabi and Cuyuna Ranges includes Animikie-Lower Keweenawan and Mesozoic-Cenozoic directions. The Animikie-Lower Keweenawan directions represent the primary remanence directions of the unaltered iron formation which have been rotated and dispersed to some extent by tectonic deformation. The deformation probably occurred during the Penokean orogeny. Therefore, it is concluded that the Biwabik and Deerwood iron formations were deposited contemporaneously with the Animikian Gunflint iron formation. The Mesozoic-Cenozoic directions represent the secondary remanence directions of the soft ore. Some of the intermediate remanence directions of the Lower Keweenawan-Cenozoic direction may represent partially altered iron formation with mixed primary and secondary remanence components (e.g., Fig. 11). This supports the hypothesis that the soft, direct-shipping iron ore deposits of the Mesabi and Cuyuna Ranges were formed by weathering primary iron formation. The groundwater solutions

oxidized the primary iron minerals to form hematite (martite)-goethite-psilomelane with a secondary CRM and left them as residual iron ore deposits after leaching the siliceous gangue. The absence of Mesozoic-Cenozoic igneous rocks in the Lake Superior region militates against the acceptance of the hydrothermal hypothesis for the ore and its remanence. This study therefore supports the interpretation of ore genesis proposed by Royce (33) in 1942 for these deposits.

7. The hematite-psilomelane deposits of the Sultana-Hopkins, Mangan-Stai, and Mahnomen No. 1 mines were formed at the same time as the goethite-limonite deposits of the Sagamore Mine. The difference in mineralogy between the mines is attributed to two different facies of primary iron formation. The two facies were most probably contemporaneous in age of deposition as postulated by Schmidt and Dutton (36).

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REFERENCES

1. As, J. A., 1960, Instruments and measuring methods in paleomagnetic research: Med. en Verh. Kon. Nederl. Meteor. Inst., De Bilt, no. 78, 56 pp.
2. Carmichael, C. M., 1959, Remanent magnetism of the Allard Lake ilmenites: Nature v. 183, p. 1329.
3. Cox, A., and Doell, R. R., 1960, Review of paleomagnetism: Geol. Soc. Amer. Bull. 71, p. 645–768.
4. Dubois, P. M., 1962, Paleomagnetism and the correlation of Keweenawan rocks: Geol. Surv. Canada Bull. 71, 75 pp.
5. Dubois, R. L., 1962, Magnetic characteristics of a massive hematite body: J. Geophys. Res., v. 67, p. 2887–2893.
6. Fisher, R. A., 1953, Dispersion on a sphere: Proc. Roy. Soc. London A., v. 217, p. 295–305.
7. Gill, J. E., 1924, Gunflint iron-bearing formation: Geol. Surv. of Canada, Sum. Rept. C., p. 28–88.
8. Goldich, S. S., Nier, A. O., Baadsgaard, H., Hoffman, J. H., and Krueger, M. W., 1961, The Precambrian geology and geochronology of Minnesota: Minn. Geol. Surv. Bull. 41, 193 pp.

9. Goodwin, A. M., 1956, Facies relations in the Gunflint iron formation: ECON. GEOL., v. 51, p. 565–595.
10. Gross, W. H., and Strangway, D. W., 1961, Remanent magnetism and the origin of hard hematites in Precambrian banded iron formation: ECON. GEOL., v. 56, p. 1345–1362.
11. Grout, F. F., and Broderick, T. F., 1919, The magnetic deposits of the eastern Mesabi Range, Minnesota: Minn. Geol. Surv. Bull. 17, 58 pp.
12. ——, and Wolff, J. E., Sr., 1955, The geology of the Cuyuna district, Minnesota, a progress report: Minn. Geol. Surv. Bull. 36, 144 pp.
13. Gruner, J. W., 1930, Hydrothermal oxidation and leaching experiments: their bearing on the origin of Lake Superior hematite-limonite ores: ECON. GEOL., v. 25, p. 697–719 and p. 837–867.
14. ——, 1937, Hydrothermal leaching of iron ores of the Lake Superior type—a modified theory: ECON. GEOL., v. 32, p. 121–130.
15. ——, 1946, The mineralogy and geology of the taconites and iron ores of the Mesabi range, Minnesota: Iron Range Resources and Rehabilitation Commission, St. Paul, Minn., 127 pp.
16. ——, 1956, The Mesabi range: Geol. Soc. America Guidebook for Field Trips No. 1-1956, p. 182–215.
17. Harder, E. C., and Johnston, A. W., 1918, Preliminary report on the geology of east-central Minnesota including the Cuyuna iron ore district: Minn. Geol. Surv. Bull. 15, 178 pp.
18. Hargraves, R. B., 1959, Magnetic anisotropy and remanent magnetism in hemo-ilmenite from ore deposits at Allard Lake: J. Geophys. Res., v. 64, p. 1565–1578.
19. Howell, L. G., and Martinez, J. D., and Stratham, E. H., 1958, Some observations on rock magnetism: Geophysics, v. 23, p. 285–298.
20. Irving, E., 1964, Paleomagnetism: John Wiley & Sons, Inc., New York, 399 pp.
21. James, H. L., 1954, Sedimentary facies of iron formation: ECON. GEOL., v. 49, p. 235–293.
22. ——, 1955, Zones of regional metamorphism in the Precambrian of northern Michigan: Geol. Soc. Am. Bull. 66, p. 1455–1488.
23. Leech, J. B., Lowden, J. A., Stockwell, C. H., and Wanless, R. K., 1963, Age determinations and geological studies: Geol. Surv. Canada Paper 63-17, 140 pp.
24. Leith, C. K., 1903, The Mesabi iron-bearing district of Minnesota: U. S. Geol. Surv. Mon. 42, 316 pp.
25. ——, Lund, R. C., and Leith, A., 1935, Precambrian rocks of the Lake Superior region: U. S. Geol. Surv. Prof. Paper 184, 34 pp.
26. Lepp, H., and Goldich, S. S., 1964, Origin of the Precambrian iron formation: ECON. GEOL., v. 59, p. 1025–1060.
27. Lowden, J. A., 1961, Age determinations by the Geological Survey of Canada, Rept. 2 on isotopic ages: Geol. Surv. Canada Paper 61-17, 127 pp.
28. ——, Stockwell, C. H., Tipper, H. W., and Wanless, R. K., 1963, Age determinations and geological studies: Geol. Surv. Canada Paper 62-17, 140 pp.
29. Marsden, R. W., 1956, Northeastern Minnesota and the Lake Superior Precambrian correlations: Geol. Soc. America, Guidebook for Field Trips No. 1—1956, p. 20–28.
30. Moorhouse, W. W., 1960, Gunflint Iron Range in the vicinity of Port Arthur: Ontario Dept. of Mines, 69-7, p. 1–40.
31. ——, and Beales, F. W., 1962, Fossils of the Animikie, Port Arthur, Ontario: Trans. Roy. Soc. Canada, 56-111, p. 97-110.
32. Nairn, A. E. M., 1960, Paleomagnetic results from Europe: J. Geol., v. 68, p. 285–306.
33. Royce, S., 1942, Lake Superior iron deposits: *in* W. H. Newhouse, 1942, Ore deposits as related to structural features: Princeton Press, Princeton, New Jersey, p. 54–63.
34. Schmidt, R. G., 1958, Bedrock geology of the southwestern part of the North Range, Cuyuna district, Minnesota: U. S. Geol. Survey Min. Inv. Field Studies Map MF-181, 3 sheets.
35. ——, 1959, Bedrock geology of the northern and eastern parts of the North Range, Cuyuna district, Minnesota: U. S. Geol. Survey Min. Inv. Field Studies Map MF-182, 5 sheets.
36. ——, and Dutton, G. E., 1957, Bedrock geology of the south-central part of the North Range, Cuyuna district, Minnesota: U. S. Geol. Survey Min. Inv. Field Studies Map MF-99, 3 sheets.
37. Stockwell, C. H., 1964, Fourth report on structural provinces, orogenies, and time-classification of the Canadian Precambrian Shield: Geol. Surv. Canada Paper 64-17 (pt. II), p. 1–26.
38. Strangway, D. W., 1964, Rock magnetism and dike classification: J. Geol., v. 72, p 648–663.
39. Symons, D. T. A., 1965, Paleomagnetic studies of Lake Superior iron ore deposits: Ph.D. Thesis, University of Toronto, Canada, 185 pp.

40. Tanton, T. L., 1931, Fort William and Port Arthur, and Thunder Cape map area, Thunder Bay district, Ontario: Geol. Surv. of Canada Mem. 167, 222 pp.
41. Tyler, S. A., 1949, Development of Lake Superior soft iron ores from metamorphosed iron formation: Bull. Geol. Soc. Am., v. 60, p. 1101–1124.
42. Van Hise, C. R., and Leith, C. I., 1911, Geology of the Lake Superior region: U. S. Geol. Survey Mon. 52, 641 pp.
43. Vlassov, A. Ya, and Aparin, V. P., 1963, On the late Precambrian paleomagnetism in Sinian deposits of the Yenisei Ridge: Akad. Nauk. SSSR Izv. Geophys. Ser., p. 451–454 (English Translation p. 278–280).
44. White, D. A., 1954, The stratigraphy and structure of the Mesabi Range, Minnesota: Minn. Geol. Surv. Bull. 38, 89 pp.
45. Winchell, H. V., 1892, The Mesabi iron range: Minnesota Geol. Surv. Ann. Rept. 20, p. 111–180.
46. Wolff, J. E., 1915, Ore bodies of the Mesabi range: Eng. Min. Jl., v. 100, p. 89–94, 135–139, 178–185, 219–224.
47. Zapffe, C., 1928, Geologic structure of the Cuyuna iron district, Minnesota: ECON. GEOL., v. 23, p. 612–646.
48. Zavoiski, V. N., and Krutikhovskaya, Z. A., 1961, The remanent magnetization of ferruginous quartzites in the southern closure of the Krivoi Rog Synclinorium: Akad. Nauk. SSSR Izv. Geophys. Ser., p. 1150–1157 (English Translation p. 765–769).

APPENDIX A

The following descriptions of the mineralogy are from the examination of the samples in polished sections which were subjected to alternating field demagnetization.

a) Gunflint iron formation, sample T-22 from the Lower member. The polished section contains one percent metallics in the form of fine-grained euhedral crystals of pyrite, magnetite, hematite and chalcopyrite in decreasing order of abundance.

b) Gunflint iron formation, sample T-8 from the Upper member. The polished section contains one percent metallics in the form of magnetite which occurs as disseminated grains and as fine particles in the silicate granules.

c) Biwabik iron formation, sample P-14 from the Lower Cherty member. The hematite occurs as coarse euhedral crystals pseudomorphic after magnetite and as irregular clusters surrounding medium-grained silicate granules. Trace amounts of unaltered magnetite were noted.

d) Biwabik iron formation, sample P-5 from the Lower Slaty member. Magnetite occurs as massive irregular blebs within quartz gangue and it contains trace amounts of hematite as inclusions. The magnetite is distinctly confined to layers.

e) Biwabik iron formation, sample P-9 from the Upper Cherty member. Hematite occurs in layers within the gangue and in the form of irregularly-shaped grains and blebs.

f) Deerwood iron formation, sample J-3 from a slate unit. Magnetite occurs as fine grained euhedral crystals interlaminated with gangue. The magnetite is partially altered to hematite. There are a few grains of goethite.

g) Deerwood iron formation, sample J-5 from a quartzite unit. Hematite occurs as irregularly shaped grains which are disseminated through the gangue in preferential laminations.

h) Deerwood iron formation, sample J-21 from soft ore within the iron formation. Primary blebs of magnetite have been mostly altered to hematite, which also occurs in massive stringers. Minor amounts of goethite coexist with the hematite.

i) Deerwood iron formation, sample K-10 from soft ore within the iron formation. Goethite occurs in irregular patches and forms the groundmass. The groundmass contains gangue inclusions and a few magnetite crystals that have been mostly altered to hematite.